# Vectors and Vector-Valued Functions

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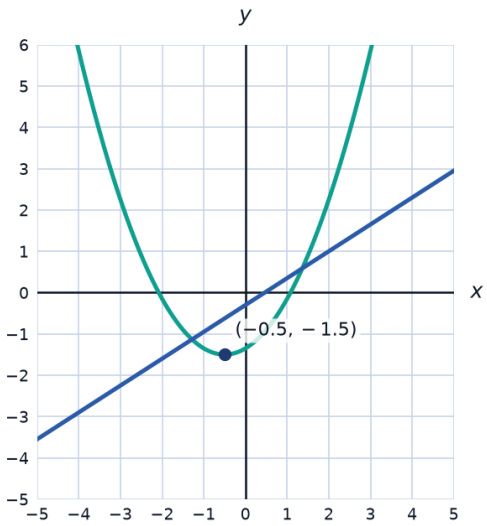
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the region $R$ in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving $R$ about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_a^b (5 - x)\sqrt{x}\,dx$. Find the value of $k$.

**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the vector $\mathbf{v}$ with initial point $P = (1, 2)$ and terminal point $Q = (4, 6)$. What are the components of $\mathbf{v}$?



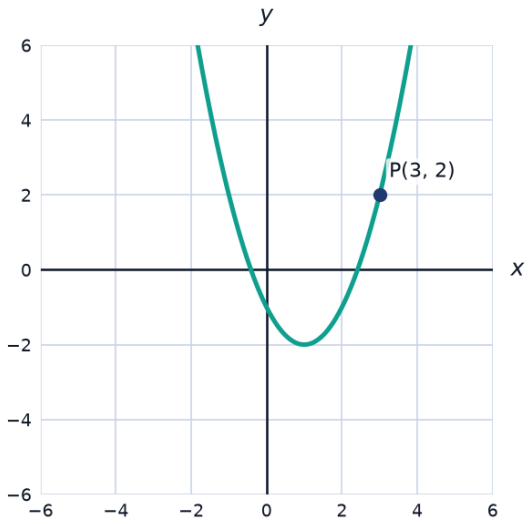

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the curvature of $\mathbf{r}(t) = \langle 3\cos(t), 3\sin(t), 3t\rangle$ at $t = \pi$.

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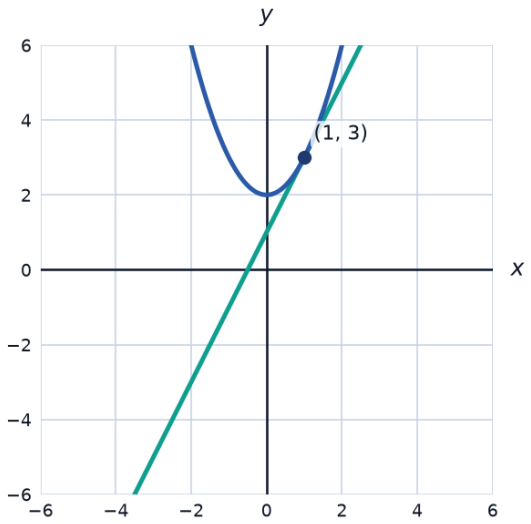
**4** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of $f$ at point $P$?




**5** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

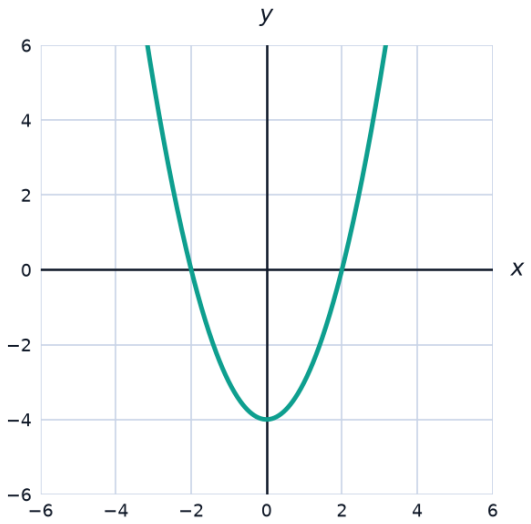
The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of $c$.




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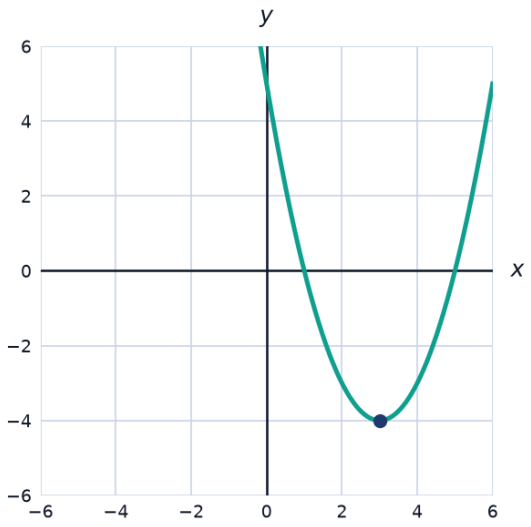
**6** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi \int_a^b (x^2 - 4)^2 dx$, find the value of $b - a$.




**7** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 6x + 5$. The function has a local minimum at which value of $x$?

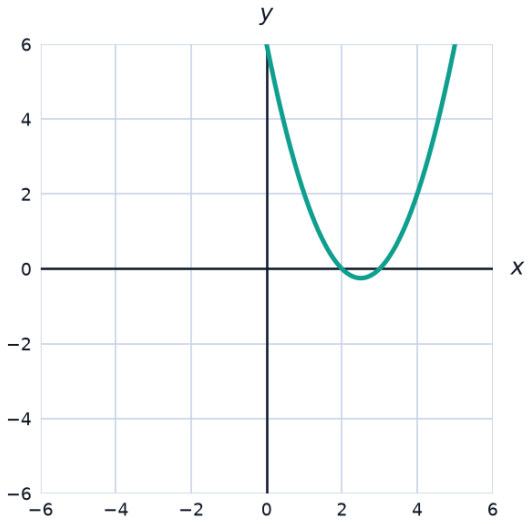



## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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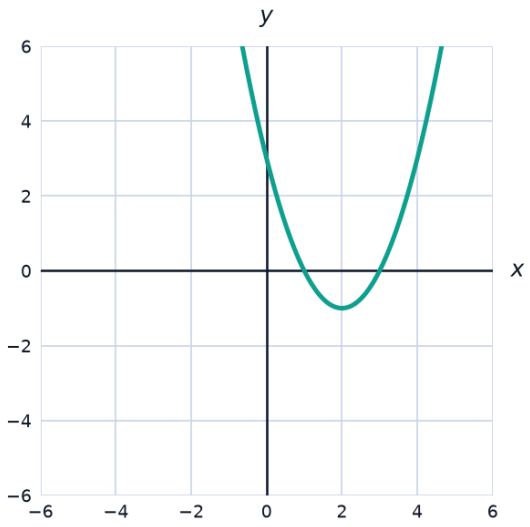
**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 5x + 6$. What is the derivative of $f(x)$ with respect to $x$?




**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

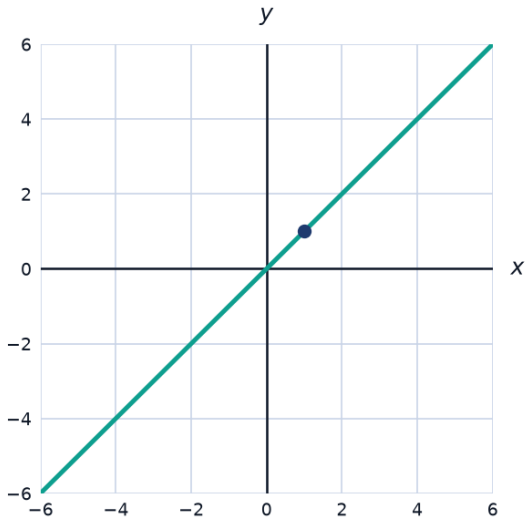
The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The function has a local minimum. What are the coordinates of the local minimum?




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**10** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as $x$ approaches 1?




**11** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

**12** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the region bounded by the curve $y = x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. What is the area of this region?

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of $f$ with respect to $y$, denoted as $\frac{\partial f}{\partial y}$?

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle's position is described by the parametric equations $x(t) = 3t + 1$ and $y(t) = 2t - 2$. What are the coordinates of the particle when $t = 3$?

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

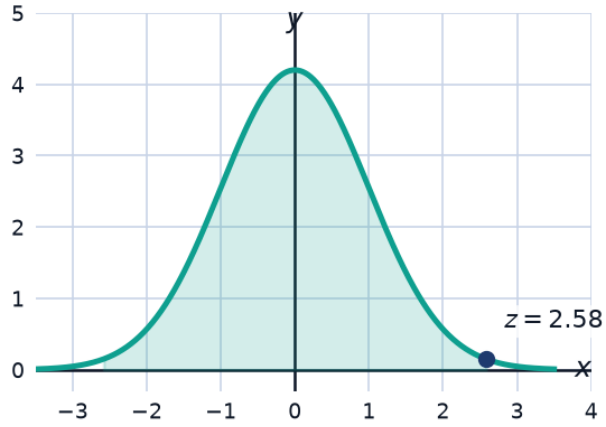
**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area Between Curves

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$. Then $\int_0^{\pi/2} \sin^6 x\,dx =$

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**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.02 Volumes of Revolution

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration $a_T$ and the normal component $a_N$ at $t = 1$.




**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Level 2 · Medium Find the area of the triangle with vertices $A(1, 1, 0)$, $B(3, 0, 2)$, and $C(0, 2, 4)$.

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Separable Differential Equations

Level 1 · Foundations Determine whether the vectors $\mathbf{u} = \langle 2, -3, 4 \rangle$ and $\mathbf{v} = \langle 6, 8, 3 \rangle$ are orthogonal, parallel, or neither.

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Precalculus Review for Calculus

[Vectors & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the cross product $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 1, 0, 2 \rangle$ and $\mathbf{v} = \langle 0, 3, -1 \rangle$.

**a.** $\langle -6, 1, 3 \rangle$

**b.** $\langle 6, -1, -3 \rangle$

**c.** $\langle -6, -1, 3 \rangle$

**d.** $\langle 6, 1, 3 \rangle$

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of the parallelogram determined by $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 3, 4 \rangle$.

**a.** $\sqrt{89}$

**b.** $9$

**c.** $\sqrt{73}$

**d.** $10$

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the unit tangent vector $\mathbf{T}(t)$ to $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$.

**a.** $\langle -\frac{3}{5}\sin t, \frac{3}{5}\cos t, \frac{4}{5} \rangle$

**b.** $\langle -3\sin t, 3\cos t, 4 \rangle$

**c.** $\langle -\frac{3}{4}\sin t, \frac{3}{4}\cos t, 1 \rangle$

**d.** $\frac{1}{5}\langle 3\cos t, 3\sin t, 4t \rangle$

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the arc length of the circular helix $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$ for $0 \le t \le 2\pi$.

**a.** $10\pi$

**b.** $5\pi$

**c.** $20\pi$

**d.** $6\pi$

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the formula for curvature $\kappa(t)$ of a space curve in terms of $\mathbf{r}'$ and $\mathbf{r}''$.

**a.** $\kappa(t) = \frac{||\mathbf{r}'(t) \times \mathbf{r}''(t)||}{||\mathbf{r}'(t)||^3}$

**b.** $\kappa(t) = \frac{||\mathbf{r}''(t)||}{||\mathbf{r}'(t)||^2}$

**c.** $\kappa(t) = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{||\mathbf{r}'(t)||^3}$

**d.** $\kappa(t) = ||\mathbf{r}''(t)||$

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**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Find the curvature of a circle of radius $R$.

**a.** $\kappa = 1/R$

**b.** $\kappa = R$

**c.** $\kappa = 2\pi R$

**d.** $\kappa = 1/R^2$

**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
If position is $\mathbf{r}(t) = \langle t^2, 4t, -16t^2 + 64t \rangle$, find the velocity and acceleration vectors at $t = 1$.

**a.** $\mathbf{v}(1) = \langle 2, 4, 32 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$

**b.** $\mathbf{v}(1) = \langle 1, 4, 48 \rangle, \mathbf{a}(1) = \langle 2, 0, -16 \rangle$

**c.** $\mathbf{v}(1) = \langle 2, 4, 0 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$

**d.** $\mathbf{v}(1) = \langle 2, 4, -32 \rangle$

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Find the speed at $t = 0$ for a particle with velocity $\mathbf{v}(t) = \langle 3\cos t, 4, 3\sin t \rangle$.

**a.** 5

**b.** 7

**c.** 4

**d.** $\sqrt{7}$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Decompose acceleration into tangential and normal components: $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$. What is $a_T$?

**a.** $a_T = \frac{\mathbf{v} \cdot \mathbf{a}}{||\mathbf{v}||}$

**b.** $a_T = \frac{||\mathbf{v} \times \mathbf{a}||}{||\mathbf{v}||}$

**c.** $a_T = ||\mathbf{a}||$

**d.** $a_T = \frac{\mathbf{a}}{||\mathbf{v}||}$

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**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the arc length of $\mathbf{r}(t) = \langle 2t, t^2, \ln(t) \rangle$ for $1 \le t \le 3$.

**a.** $2\ln(3) + 16$

**b.** $\ln(3) + 8$

**c.** $\frac{\ln(3)}{2} + 4$

**d.** $\ln(3) + 9$

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Arc length of $\mathbf{r}(t) = (e^t \cos t, e^t \sin t, e^t)$ from $t = 0$ to $t = 1$:

**a.** $e - 1$

**b.** $\sqrt{3}(e - 1)$

**c.** $\sqrt{2}(e - 1)$

**d.** $e\sqrt{3}$

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For $\mathbf{r}(t) = \langle 2t, 3t^2, t \rangle$, find the normal component of acceleration at $t = 1$.

**a.** $\frac{6\sqrt{205}}{41}$

**b.** $\frac{36\sqrt{41}}{41}$

**c.** 6

**d.** 36

**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The magnitude of cross product $|\mathbf{a} \times \mathbf{b}|$ equals the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. For $|a| = 3, |b| = 4, \theta = \pi/6$:

**a.** 24

**b.** 12

**c.** 3

**d.** 6

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Cross product $\mathbf{a} \times \mathbf{b}$ is orthogonal to:

**a.** $\mathbf{a}$ only

**b.** $\mathbf{b}$ only

**c.** neither

**d.** both $\mathbf{a}$ and $\mathbf{b}$

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**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Flux of $\mathbf{F} = (x, y, 0)$ through lateral surface of cylinder $x^2 + y^2 = 1$, $0 \le z \le 2$, outward:

**a.** $0$

**b.** $2\pi$

**c.** $4\pi$

**d.** $\pi$

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Parametric surface $\mathbf{r}(u, v) = (u\cos v, u\sin v, u^2)$. At $(u, v) = (1, 0)$:

**a.** tangent plane is $z = x^2$

**b.** tangent plane is $z = 2x - 1$

**c.** normal is $(0, 0, 1)$

**d.** surface is a sphere

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Vector triple product: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) =$

**a.** $\mathbf{0}$

**b.** $(\mathbf{a} \cdot \mathbf{b})\mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b}$

**c.** $\mathbf{b}(\mathbf{a} \cdot \mathbf{c}) + \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$

**d.** $(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Separable Differential Equations

Which is an equation of the plane through $(-3,\ 1,\ 4)$, $(-2,\ 2,\ 4)$ and $(1,\ -1,\ 1)$?

**a.** $x - y + 2z = -4$

**b.** $x + y + 2z = 6$

**c.** $x - y + 2z = 4$

**d.** $x + y = -2$

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**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find the work done by $\mathbf{F} = (x^2, y^2, z^2)$ along line segment from $(0, 0, 0)$ to $(1, 1, 1)$:

**a.** 1

**b.** $\frac{2}{3}$

**c.** 3

**d.** $\frac{1}{2}$

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Area of surface $z = f(x, y)$ over $D$:

**a.** $\iint_D f\,dA$

**b.** $\iint_D \sqrt{1 + f_x^2 + f_y^2}\,dA$

**c.** $\iint_D (f_x + f_y)dA$

**d.** $\iint_D f^2\,dA$

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Sequences and Series

For $\mathbf{r}(t) = \langle 2t, t^2, t\rangle$, find the tangential component of acceleration at $t = 2$.

**a.** $\frac{2\sqrt{105}}{21}$

**b.** $\frac{8\sqrt{21}}{21}$

**c.** 2

**d.** 8

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 12.01 Convergence Tests

The torsion $\tau$ of a space curve measures how fast the curve twists out of its osculating plane. For a helix $\mathbf{r} = (a\cos t, a\sin t, bt)$, as $b \to 0$ (curve flattens to a circle), $\tau \to$

**a.** $1/a$

**b.** 0

**c.** $\infty$

**d.** $b$

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 12.02 Power Series and Taylor Series

Find the area of the triangle with vertices $(1,\ 0,\ 2), (1,\ 1,\ 2), (-3,\ 3,\ -3)$.

**a.** $\sqrt{41}$

**b.** $\frac{\sqrt{41}}{3}$

**c.** $\frac{\sqrt{41}}{2}$

**d.** $\frac{3}{2}$

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**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

The normal vector to the surface $F(x, y, z) = 0$ at a point is:

**a.** $(F_x, F_y, F_z) = ablaF$

**b.** $(F_{xx}, F_{yy}, F_{zz})$

**c.** the Hessian of $F$

**d.** $(F_y, F_x, 0)$

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Double Integrals

Distance from point $P = (1, 2, 3)$ to plane $2x + y - 2z = 5$:

**a.** $\frac{7}{3}$

**b.** $\frac{2}{3}$

**c.** $1$

**d.** $3$

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